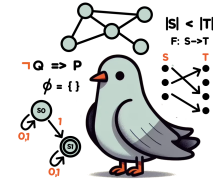


CS 2800 Mathematical Foundations of Computing

Fall 2025 Syllabus



General Information

Brief description: This course covers the mathematics that underlies most of computer science. Topics include mathematical induction; logical proof; propositional and predicate calculus; sets, functions, and relations; graphs; combinatorics and discrete mathematics; basic probability theory; and finite-state machines. These topics are discussed in the context of applications to many areas of computer science.

Credit Hours: This is a 4-credit course, which can only be taken for a letter grade.

Course Website: <https://canvas.cornell.edu/courses/77893>

Course Staff:

Instructors:

- Anke van Zuylen, Gates Hall 447.
- Mark Moeller, Gates Hall (office TBA).

Course Admin: Corey Torres, Gates Hall 401.

Teaching Assistants: We are lucky to have 40 TAs for the course. You can find more information about them, on Canvas.

Contact information:

- **Office hours:** See schedule on Canvas for the office hours for all instructors, TAs and consultants.
- **Ed Discussion:** Use our Ed Discussion site to ask questions about the material, including lectures, homework, and exams, as well as general questions about course organization and logistics. Please don't forget to search whether a question has been asked already!
- **Email:** To email the instructors or the course administrator about personal matters, please use cs2800-prof@cornell.edu.
- **1-on-1 Appointment:** You can make an appointment with professor Van Zuylen using calendly. Appointments slots are limited, and are meant for discussing topics like personal circumstances that you cannot discuss in public office hours (so not for homework questions!)

Time and Location:

- *Lectures* are held MWF 11:15am-12:05am, in the Statler Auditorium (Statler 185) from August 25 until December 8 (with the exception of university-scheduled breaks).
- *Discussion sections* are 50-minute meetings, held at different times and locations on WThF, in which you will take a 10-minute quiz on the most recent homework, and work in small groups and with course TAs on a subset of the problems in the homework assignments.

Pre/corequisites:

- A calculus course at the level of AP Calculus AB or MATH 1110, is a corequisite for the course. That means that this course can be taken concurrently with CS 2800. Some level of mathematical familiarity (at the level of a high school curriculum) is assumed. You should be comfortable working with familiar functions (e.g. polynomial and exponential functions) and performing simple arithmetic without a calculator. The last part of the Probability module of the course assumes some familiarity with differential and integral calculus.
- A programming course, for example CS 1110 or CS 2110, is a corequisite for the course. That means that this course can be taken concurrently with CS 2800.

Course description

In this course, we will study the mathematical foundations of computer science starting at the most basic level. The focus of the course is on many *discrete structures* (i.e. objects that are characterized by distinct, separated values as opposed to the *continuum* of real numbers) such as logical formulas, sets, functions, relations, graphs, and automata. These topics are discussed in the context of applications to many areas of computer science. An emphasis of this course, in contrast to other mathematics classes that you may have taken, is on writing rigorous justification (or a *proof*) to fully explain your mathematical reasoning. As such, this course not only has the goal of introducing you to important definitions and theorems but also of teaching you how to think critically and gain confidence when solving new mathematical or computational problems.

Learning objectives

On completing this course, students should be able to:

- Use **logical notation** to define and reason about statements expressed in informal language.
- Write **mathematical proofs** using clear and precise reasoning; recognize when to use various proof techniques; find errors in faulty proofs.
- Precisely state induction hypotheses and write **inductive proofs**; write and use inductive definitions.

- Reason about **fundamental mathematical concepts** such as sets, functions, relations, graphs and their properties.
- Apply **combinatorial reasoning** to solve counting problems arising in real-world applications; use the pigeonhole principle to prove statements.
- Use **probability concepts and notation** such as conditional probability, Bayes' rule, random variables, and tail bounds to model and analyze real-world applications; compute associated quantities such as probabilities, expectations, variance, covariance, and correlation.
- Prove properties of **regular languages and automata**; design automata and regular expressions accepting or denoting a certain regular language; prove that a language is not regular.

Course Textbook and Other Materials

Course Text

A beautiful set of comprehensive course notes, written by Matt Eichhorn, is available on Canvas. These notes are broken down by lecture topic, and include many worked examples and additional exercises for you to gain mastery of the material.

Optional Textbooks

There are many excellent books about discrete mathematics that offer an alternative presentation of the material, which may be helpful in your studies. Some additional references that we recommend are:

- *Discrete Mathematics and its Applications* by Kenneth Rosen. Refer to this text (any edition is fine) for many great exercises with solutions.
- *Discrete Mathematics ZY Book* by Sandy Irani. Get (paid) access to this if you find it helpful to answer brief autograded questions that check your understanding while reading.
- *A course in Discrete Structures*, a concise set of lecture notes for a previous version of this course, by Rafael Pass and Wei-Lung Dustin Tseng.
- *Mathematics for Computer Science* by Eric Lehmann, Albert Meyer, and Tom Leighton.
- *Essential Discrete Mathematics for Computer Science* by Harry Lewis and Rachel Zax.

The topics covered by these texts differ (to varying degrees) from our course. Also, the levels of formality and difficulty of these texts vary, and some of the notational choices differ from or conflict with ours. For consistency, we will expect you to use the notation presented in lectures and the course notes.

Course Websites

We use Canvas for distributing handwritten lecture notes, homework solutions, and discussion assignments. You should be enrolled automatically into Canvas, but if not (for example, because your registration has not been processed yet), please contact the course administrator to get added manually. All other websites we use (Gradescope for homework submission, Ed Discussions for Q&A, etc.) are linked from the course Canvas site; you should automatically get added to the rosters for these other sites if you access them through the Canvas link.

You are expected to check the Ed Discussion site regularly for announcements, and keep an eye on the Canvas site for useful materials.

Grading

Your grade will be based on weekly homework and in-discussion quizzes, two prelims, one final exam, and participation in lecture polls and course evaluations. Each of these components is given the following weights:

- homework: 13%
- quizzes: 13%
- prelims: 40%
 - there are two prelim exams, that are weighted as follows:
 - lower prelim score: 15%
 - higher prelim score: 25%
- final exam (cumulative): 30% to 34%
- participation: 0 to 4%
 - participation scores are made up of the following components:
 - Polleverywhere: 3%
 - course evaluation (mid-semester and end-of-semester): 1%

At the end of the semester, we will compute the overall score for each individual student using the above scheme (where each participation component only counts if the score you get for the component is higher than your final exam score; otherwise, the weight is shifted to the final exam).

Homework

This course has weekly homework assignments. Each homework assignment consists of two parts: the short-answer problems (SA) are designed to help you to check your understanding and synthesize concepts, and the long-answer problems (LA) will allow you to practice solving

more complex problems, and how to write well-structured and clear mathematical arguments. If you follow the course policies (see below) and focus on learning from the homework (rather than just getting it done), the homework will give you an excellent preparation for the exams.

Homework assignments are typically posted on Wednesday morning and due the next Tuesday at 8pm. You can view a (tentative) schedule of the assignments on the Course Calendar. The (tentative) schedule contains 11 homework assignments in total.

Homework needs to be submitted on Gradescope. There will be a separate place to enter your answer to the short-answer and long-answer problems. It is OK if you handwrite your answers for the long-answer problems, and upload a photo (converted to a single pdf), but we encourage you to learn how to typeset them using LaTeX. Resources to help you get familiar with LaTeX are available on Canvas, and we post the LaTeX source files of the homework so you can quickly learn the important syntax and commands.

Quizzes

Each week you will be given a 10-minute quiz, which will be administered in your discussion section. The quiz consists of some (usually one) of the exercises of the homework that was due earlier that week.

The quiz will be administered at the start of the class, so make sure to come on time. There will be no makeup quizzes offered. Your discussion TA will grade your quizzes.

You can only take the quiz in the discussion section that you signed up for – notify the TA immediately when you change discussion sections. **Your discussion TA will not be able to upload quiz grades to your account or grade your quiz if you are registered in a different section on Gradescope.**

Exams

There will be two prelims and a final exam. The prelim exams are scheduled for Tuesday September 30, 7:30pm and Thursday November 20, 7:30pm. The date for the final exam is to be determined. The exams will be held in-person.

Exams for CS2800 are closed-book and closed-notes and do not require any equipment except pen and paper. You can expect questions on the exam to be similar to the homework problems, so a good understanding of the key ideas used in the homework problems is necessary to do well. Review materials that include questions from previous years' exams will be provided a week before the exam.

Exams for students with SDS accommodations are administered by SDS. Please see the Accommodations section later in this syllabus for detailed instructions.

Polleverywhere

Lecture participation is measured using PollEverywhere; there will be approximately 1-4 polls per lecture, and you get credit for each poll you complete (regardless of whether your answer is correct).

Course Policies

Allowances for absences and minor mistakes

To help you balance your different responsibilities, and to reduce some of the pressure of the weekly grind of the course, we have the following policies:

- We drop the 3 lowest scores for the weekly deliverables (homework assignments and quizzes). Depending on what is best in your situation, we will either drop the 3 lowest homework scores, or 2 homework scores and a 1 quiz score, or 1 homework score and 2 quiz scores, or 3 quiz scores. If you are not able to attend your discussion section and miss the quiz, the quiz score of 0 is eligible to be dropped, similarly, a homework that is not submitted gets a score of 9 that is eligible to be dropped.
- Your score for each homework gets multiplied by 10/8.5 (rounded down to 100 if necessary). In other words, you only need to score 85% for full credit.
- Your overall score for polleverywhere get multiplied by 10/8.5. In other words, you can miss up to 15% of the lecture polls without penalty.

Submission deadlines

Homework deadlines are at 8pm. You can submit the assignments up to 12 hours late. There is no penalty for late submission. Because solutions are posted at 8am on the day following the deadline, to allow students sufficient time to review them before the quiz, we cannot accept homework after this time for any reason.

Collaboration

Collaboration and conversation is an important part of how ideas get generated. We encourage you to discuss with a small group of your peers in the course to brainstorm ideas for how to solve the assigned exercises. However, **your solution must be written up completely on your own.** So feel free to talk as long as necessary to get a firm grasp of how to solve the exercises, but do not take notes of your discussion, and do not share digital or written notes or images of your work in any form with each other.

Just like in research, you must include a collaborator statement on each homework assignment, including a list of all students with whom you discussed the homework.

Admissible Resources

The only resources that you may use for your homework are course materials that are linked on Canvas (the lecture notes/slides, discussion worksheets, and past assignments) and the optional texts listed in the syllabus. Do not consult outside websites to help prepare your solutions. **Using**

other resources, including ChatGPT and similar chatbots, to write the answers to the homework assignments is not allowed because it defeats the purpose of the homework, which is for you to get feedback on your own understanding and writing. If you feel the resources available to you are insufficient, talk to course staff or ask questions on Ed.

Regrades

Regrade requests for homework assignments and exams should be submitted on Gradescope. Regrade requests are accepted for four days, starting one day after grades are released. We may regrade the entire exam or problem set, and your score may go up or down. When submitting a regrade request, be specific about which rubric items were misapplied, and how you feel the grading should be corrected. Point to specific parts of your answer.

Academic integrity

Each student in this course is expected to abide by the Cornell University Code of Academic Integrity. Any work submitted by a student in this course for academic credit must be the student's own work. The complete code is available at <https://deanoffaculty.cornell.edu/faculty-and-academic-affairs/academic-integrity/code-of-academic-integrity/>. This course is participating in Accepting Responsibility (AR), which is a pilot supplement to the Cornell Code of Academic Integrity (AI). For details about the AR process and how it supplements the AI Code, see the AR website.

Submitting homework solutions that are not your own work or that do not abide by the course policies (for example, submitting close-to-identical work as a classmate, or work that has clear signs of the use of inadmissible resources like ChatGPT) is a violation of academic integrity. Similarly, accessing unauthorized resources or communicating with any person during a quiz violates the code of academic integrity. Such academic dishonesty undermines the educational process by preventing your own learning and skill development, compromising the integrity of assessment, and devaluing the achievements of honest students. The penalty for such violations is a reduction of the score for the impacted assignment to 0, as well as a reduction in the number of dropped deliverables by 1.

Communicating with any person or accessing unauthorized resources during an exam for CS2800 results in an F in the course.

Course Material Copyright

Course materials posted on Canvas, gradescope, or Ed Discussions are intellectual property belonging to the author. Students are not permitted to buy or sell any course materials without the express permission of the instructor. Such unauthorized behavior constitutes academic misconduct.

Advice for Success

Here are a few tips for succeeding in this course:

- **Use the homework to make mistakes and learn!** We see many students each semester who do great on the homework, and who realize midsemester (often during a prelim) that they have no idea how to solve, or even read, the problems on their own. Very often, these students have been relying too much on their collaborators or TAs or disallowed resources to solve the homework problems. On the other side of the coin, we also see many students who don't do that well on the homework, and who end up with an A in the course. So please take advantage of the fact that a single assignment counts for only 1–2% of your grade and allow yourself to make your own mistakes. When you do the homework the right way, you get the best possible preparation for doing well on an exam!
- **Do not memorize proofs.** Prepare for the in-discussion quizzes and exams by making sure you understand the homework problems, and be ready to explain any answer in detail. We hope to see you write carefully phrased arguments and proofs that are your own, not ours.
- **Start your assignments early.** Even if you aren't writing anything down yet, looking over the problem set well in advance of the due date can ensure you have enough time to brainstorm possible solutions, and to clear up confusion about how to interpret a problem. Solving homework problems requires thought and creativity, and this doesn't work well on a deadline.
- **Use your discussion section to gain confidence and get feedback.** The exercises assigned each week to be solved during your discussion section give you the opportunity to work intensively with your peers and the TAs on solving the problems.
- **Talk with classmates at a similar level about ideas for the homework.** As previously stated, while you cannot share physical or digital solutions of any kind to these problems, we actively encourage you to talk to classmates while you work through them. In particular, we recommend finding a group of students to meet with throughout the semester to talk about ideas. For best results, make sure those students are at the same level of understanding of the material as you; talking through your ideas with colleagues with a similar level of understanding will make talking through ideas with each other easier and more equitable, and is more likely to leave you prepared for course exams.
- **Ask questions in class, in discussions, in office hours, and on Ed Discussions.** The material in this class moves quickly and is often cumulative. In addition, in mathematics, you need to understand every single symbol of a statement in order to make sure you are not considering a completely different problem. This takes some getting used to! So if you find yourself scratching your head after a lecture, even after consulting the textbook and course notes, you're certainly not alone, and it's better to seek help than to wait until you are more confused.

- **Don't panic if you're stuck and confused.** It is normal to struggle, and to be stuck sometimes and have no idea what to do! Struggling with the material helps you get a deeper understanding, so rather than panic, take a break, try again, and find help if you need it!

Academic Excellence Workshop ENGRG 1028

Academic Excellence Workshop (AEW) sections are available to be taken in conjunction with this course. AEWs are optional 1-credit supplemental courses which meet for one 2-hour collaborative problem-solving session each week throughout the semester. Designed to enhance student understanding, the workshops feature group work on problems at or above the level of course instruction. In the workshops, small-group problem-solving is directed by undergraduate peer educators called facilitators. The AEWs are graded S/U, based on attendance.

For this course there are multiple sections of ENGRG 1028 available, please see student center for details. You can enroll online during the add period. Space may fill up quickly – if there are no spots available in a section that fits your schedule, use the link included with the course listing in the registration system to indicate your interest and availability. For more information about AEWs, visit <http://www.engineering.cornell.edu/aew>.

Inclusiveness, Accessibility, and Support Services

Inclusiveness

We understand that our students and staff represent a rich variety of backgrounds and perspectives. Cornell University is committed to providing an atmosphere for learning that respects diversity. We expect students to communicate in a respectful manner with the instructors, course staff, and fellow students, in a way that honors the unique experiences, values, and beliefs represented by different members of our community.

You should expect and demand to be treated by your classmates and the course staff with respect. You belong here, and we are here to help you learn and enjoy this course. If any incident occurs that challenges this commitment to a supportive and inclusive environment, please let the instructor or a TA know so that we can address the issue. We are personally committed to this, and subscribe to the Computer Science Department's Values of Inclusion.

What to do if you get sick

If you are sick, your first priority should be getting better; that means getting a lot of rest, drinking fluids, going to see a doctor if needed, etc. If you feel well enough to come to lecture or discussion, please make sure to wear a mask so you don't infect others.

Zoom recordings of lectures are not available for absences, including absences due to illness. For any illness, you are expected to keep up with course material by working with a peer in the course and/or accessing the lecture notes on canvas. You are allowed to miss 15% of the lectures

and we drop up to three quizzes (see Course Policies). We do not excuse additional absences for any reason, except in the case of long term illness.

Getting support

We urge you to talk to the instructor or any of the TAs if you have any concerns about your learning or progress in the course, if you are sick for more than a week, or if you have other personal difficulties. Resources at Cornell are also available (caringcommunity.cornell.edu).

In case you experience major unexpected events (for instance, you being hospitalized unexpectedly, death of a close family member), or you have other personal concerns you want to discuss, please set up a meeting using calendly so we can figure out how to handle the situation together.

We cannot accommodate requests for exceptions for minor issues (leaving early for a break, forgetting to submit a homework, for example) — the course policies and grading scheme have been set up to take care of this.

Accommodations

This course complies with the university policies and equal access laws, and we provide accommodations for disability, religious observance, Title IX, varsity athletes, medical emergencies, and military service. Requests for academic accommodations should be made during no later than the drop deadline, except for unusual circumstances, so arrangements can be made as soon as possible. Testing-related academic accommodations are organized by SDS; please carefully read the next subsection for details. If you require accommodations that are not testing related, please email the course admin to discuss the logistics of your accommodation(s).

Testing accommodations

This course is participating in the SDS Alternative Testing Program (ATP) for the Spring 2025 semester. All exams will be centrally managed and supported by the ATP Testing Coordinator in the Office of Student Disability Services (SDS) and accessible in your SDS student portal. Exam logistics will be communicated to you via email from sds-testing@cornell.edu and accessible in your SDS student portal. The ATP will automatically send an email with the exam date, time, location 10 days before the exam, and a reminder email 48 hours prior to the exam date.

Learning Outcomes

The following is a list of skills that you will have after taking this course. This list is not meant to be exhaustive, and some topics may need to be reduced or omitted based on our pacing during the semester. A lot of the outcomes use terminology that we will introduce during the course, so the list is most useful in retrospect.

Module 1: Foundations

1. Explain the syntax and semantics of operations in propositional logic
2. Represent English statements using logical symbols
3. Use truth tables to determine whether two compound propositions are logically equivalent
4. Describe sets using set builder notation
5. Demonstrate understanding of predicates with one or more variables
6. Identify the free and bound variables in a quantified logical statement
7. Establish the truth value of statements containing one or more quantifiers
8. Demonstrate understanding of the set intersection, union, difference, Cartesian product, and power set operations
9. Write formal arguments of set containment and equality
10. Explain and carry out direct proofs, proofs by case analysis, proofs by contradiction, and proofs by contraposition
11. Construct counterexamples that refute a proposition
12. Identify and correct flaws in erroneous proofs

Module 2: Induction

13. Explain the Principle of Induction
14. Identify the steps of a proof by induction
15. Use induction to prove statements quantified over the natural numbers
16. Describe the connection between induction, recursion, and recurrences
17. Use induction to verify the closed form of a recurrence relation
18. Explain why we must sometimes strengthen an inductive statement

19. Compare and contrast ordinary and strong induction
20. Carry out inductive proofs that require multiple base cases, multiple inductive steps, and/or the strong inductive hypothesis

Module 3: Graphs and Relations

21. Translate between a visual and a formal set description of directed and undirected graphs
22. Demonstrate understanding of common graph terminology
23. Identify paths and cycles in a graph
24. Determine whether a graph satisfies common properties such as acyclicity, bipartiteness, planarity, and regularity
25. Use structural induction to establish a property of a family of graphs
26. Use structural induction to prove propositions involving graphs
27. Identify pairs of elements that satisfy a binary relation
28. Decide whether a binary relation is reflexive, symmetric, anti-symmetric, and/or transitive
29. Construct a relation on a given set that has one or more given properties
30. Justify whether or not a relation is an equivalence relation
31. Find the equivalence classes of an equivalence relation
32. Describe the quotient of an equivalence relation
33. Justify whether or not a relation is a partial order

Module 4: Functions and Cardinality

34. Invoke the formal definition to determine whether a function is well-defined
35. Compute the composition of two functions
36. Give a rigorous argument that a function is injective, surjective, or bijective
37. Describe the relationship between bijectivity and function inverses
38. Write formal proofs involving function composition, injectivity, and inverses
39. State and apply the Pigeonhole Principle

40. Indicate when one proposition is a generalization or specialization of another
41. Use the Pigeonhole Principle to establish properties of graphs
42. Explain how one can use injective functions to define the cardinalities of sets
43. Determine whether a given set is countable or uncountable
44. Write proofs by diagonalization

Module 5: Combinatorics

45. Describe multi-step processes for enumerating the items in a set
46. Compute set sizes using the Multiplication, Addition, Subtraction, Bijection, and Division rules in combinatorics
47. Prove identities using combinatorial reasoning
48. Explain and apply the Binomial theorem
49. Apply the Inclusion-Exclusion rule to calculate the size of a set
50. Differentiate between arrangements with distinguishable/indistinguishable balls/bins
51. Use the stars and stripes formula to calculate the size of a set
52. Identify how two sides of an equation count the same set, and write a combinatorial proof of the equation

Module 6: Probability and Statistics

53. Identify the components of a probability space
54. Utilize techniques from combinatorics to compute probabilities in discrete probability spaces
55. Calculate the probability of an event conditioned on another event
56. Describe how conditioning on an event changes a probability space
57. Determine whether events are independent
58. Explain and apply the Law of Total Probability
59. Use Bayes' Rule to calculate posterior probabilities
60. Describe how to use Bayesian Inference to update a prior probability based on new evidence

61. Explain how the probability mass function and Cumulative Distribution Function describe the distribution of a discrete random variable
62. Identify the pmf, CDF, expectation, and variance of discrete uniform, Bernoulli, binomial, and geometric random variables
63. Use the definition to compute the expectation of a random variable
64. Calculate and interpret the variance of a random variable
65. Demonstrate understanding of joint, marginal, and conditional distributions of random variables
66. Describe when two random variables are independent and how this can simplify calculations
67. Explain and apply the linearity of expectation
68. Calculate and interpret the covariance and correlation of two random variables
69. Use Markov's, Chebyshev's, and Hoeffding's Inequalities to calculate tail probability bounds
70. Describe when one probability inequality may be preferable to another
71. Explain the law of large numbers
72. Interpret the probability density function of a continuous random variable
73. Identify three families of continuous distributions and describe their parameters
74. Calculate statistics of continuous random variables
75. Explain the central limit theorem
76. Use maximum likelihood estimation to estimate the parameters of a distribution

Module 7: Languages and Automata

77. Differentiate between alphabets, strings, and languages
78. Give a formal description of an automaton from its graph representation
79. Determine whether a given word is in the language of a given automaton
80. Draw a DFA that recognizes a given language
81. Describe the relationship between the Pigeonhole Principle and the Pumping Lemma or Explain the Myhill-Nerode Theorem

82. Use the Pumping Lemma to argue that a language cannot be recognized by a DFA or Apply the Myhill-Nerode theorem to argue that a language is not regular
83. Draw an NFA that recognizes a given language
84. Describe the language of a given NFA
85. Perform the subset construction to convert an NFA into a DFA that recognizes the same language
86. Construct a regular expression that represents a given language
87. Apply Thompson's construction to construct an NFA that recognizes the same language as a regular expression
88. Use the state elimination procedure to write a regular expression that represents the language of an automaton
89. Apply Kleene's theorem to reason about properties of regular languages